

How Developers Think: A Persona Playbook

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Introduction

Developers are at the heart of the digital revolution, and the driving force behind the apps and systems we rely on every day.

Getting into their mindset and understanding how they think is crucial.

You can better figure out your audience's needs and preferences, communicate with them more efficiently, build trust, inform your product development, grow your developer community, improve the user experience, and so much more.

However, this is still a challenge for many. Developers are people, not a monolith, and there can be a huge difference in mindset depending on factors like level of experience, area of specialty, and personality.

This playbook is a guide for everyone wanting to **define and understand their developer audience**. Whether you're new to the technical product scene or a seasoned veteran, get into the minds of your users and create products they love with our guide to developer personas.

Chapter 1

What are personas?

What are personas?

In today's crowded marketplace, understanding your audience is more important than ever. Consumers have many choices at their fingertips, so they no longer respond to generic or one-size-fits-all strategies.

They want personalization, relevance, and to be understood. This is where personas can play a huge role.

Origin of 'personas'

In 1999, the term 'personas' was introduced publicly by Alan Cooper, a software designer and programmer from the US, in his book *The Inmates Are Running the Asylum*. However, before that, in 1983, Cooper used personas as a role-playing strategy while working on new software.

Through personas, he was able to empathize with potential customers, as well as understand their mindset better.

Cooper developed the concept over time and, over the years, it gained popularity in the software industry due to the power of personas.

What are personas?

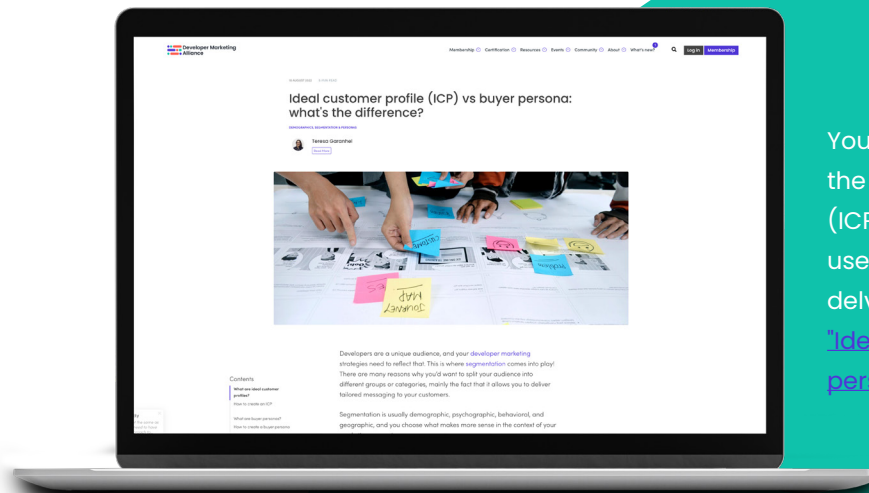
How do you define personas?

So, what is a persona?

Personas are archetypical and semi-fictional representations of a larger demographic. In this context, those who use or purchase your product.

These personas encompass a variety of information, from job titles and primary duties to goals and technical proficiency. By maintaining a human focus in every decision you make and every marketing strategy you adopt, personas help in improving your product.

A persona symbolizes your perfect customer; so, think about who your product's users or buyers are.



You might also be familiar with the term "ideal customer profiles" (ICPs), which are distinct from user and buyer personas. We delved into this topic in our ["Ideal customer profiles vs buyer personas"](#) article.

What are personas?



“Developers wield more decision-making power and influence than ever before when it comes to selecting the solutions their company will be working with. With their deep understanding of technical requirements, efficiency, and scalability, developers play a pivotal role in shaping the technological direction of their organizations.

“Even if they are not the ultimate decision-makers, their influence cannot be underestimated. Their opinions and recommendations heavily influence the choices made by key stakeholders and decision-makers.

“As trusted advisors and influencers, developers possess the ability to sway opinions, advocate for innovative tools and technologies, and champion the adoption of solutions that align with their company’s goals and objectives.”

Alice Malby, Lead Generation Manager at Front-Commerce

Keep in mind:

- Developers have a growing influence on technology-buying decisions.
- Traditional marketing approaches often fail to resonate with developers.
- Developer communities have distinct characteristics, which marketers need to consider in their marketing efforts.
- Open-source culture plays a significant role in developer communities.
- Researching and understanding the right developer community for your product is essential for effective marketing.
- Prioritize, the pain points you’re addressing, etc.

Chapter 2

User vs. buyer personas vs. ICPs

What are personas?

User and buyer personas represent people, while ICPs represent organizations.

All three of these terms are used in marketing to help businesses better understand and target their customers. In this case, the developer persona!

What are these personas?

User personas

Fictional characters created to represent a type of user that might use your product in a similar way. They also help teams to understand and empathize with their product's end users.

Buyer personas

Similar to user personas but used more in marketing and sales, not in product development. They represent your product's ideal customers and are usually based on market research and real data about current customers.

Ideal customer profiles (ICPs)

This term is used in sales and marketing to describe the type of company that would be your product's perfect customer. ICPs are often used in B2B.

The boundaries between these categories can sometimes blur, and the exact definitions can vary depending on the context or the company's specific practices.

What are personas?

Differences and similarities

But, what are the differences and similarities between user and buyer personas, and ICPs (ideal customer profiles)? Here's a handy table to help:

	User personas	Buyer personas	Ideal customer profiles
Purpose	Understand how users interact with a product/service.	Understand how customers make purchasing decisions.	Identify the perfect customer at the organizational level.
Focus	Individual user behavior and experience.	Individual buyer behavior and decision-making process.	Organization's characteristics and buying behavior.
Typical Use	UX/UI design, product development.	Marketing and sales.	B2B sales and marketing.
Based On	User research, often includes actual data from user behavior.	Market research, data about existing customers.	Market research, data about ideal customer companies.
Includes	Demographics, behavior patterns, motivations, goals.	Demographics, behavior patterns, motivations, goals, purchasing decisions.	Demographics (industry, company size, location), behavior, technographic details.

Chapter 3

The developer persona

The developer persona

Developer personas are strategic tools you can use to leverage your product design and development, specifically when creating products intended for developers. They embody the needs, pain points, characteristics, etc., of your target audience and are built with user research and real data.

What are developer personas?

Developer personas encompass a variety of details that provide a comprehensive understanding of your audience.

These details include:

- **Demographic** data, such as education and experience level.
- **Professional** information like job role, tasks, preferred programming languages, and types of projects.
- **Psychographic** aspects such as motivations, frustrations, and work habits.
- **Behavioral** traits, including problem-solving approaches and learning preferences.

The primary purpose of developer personas is to guide decision-making across product development, user experience and user interface (UX/UI) design, documentation, support, and even marketing.

By offering a focused understanding of the target users, developer personas help product teams create solutions that are better tailored to meet user needs.

This not only enhances user satisfaction but also encourages higher adoption rates, ultimately leading to the product's success.

The developer persona

What makes developers different from other personas?

It's important to mention that developers are considered a unique audience with unique skills and mindsets that set them apart from other personas.

Some key differences include:

Technical skills

Developers have a deep understanding of programming languages, data structures, algorithms, and software design principles. They can use these skills to create software applications, design databases, and solve complex technical problems.



Problem-solving mindset

Developers often approach problems with a logical, analytical mindset. They're good at breaking down complex problems into smaller, manageable parts, and they use this skill to develop solutions to technical challenges.



The developer persona

Continuous learning

The field of tech is constantly evolving, with new programming languages, products, and methodologies emerging all the time. As a result, developers are often lifelong learners who are comfortable with change and excited by the opportunity to learn new things.



Detail-oriented

Developers need to pay close attention to detail. A small error in a line of code can lead to significant problems in a software application, so developers need to be meticulous in their work.



Creativity

While it may not seem like it at first glance, programming is a highly creative endeavor. Developers often need to come up with innovative solutions to complex problems, and they have a lot of freedom in how they design and implement those solutions.



Collaboration

Most software development is a highly collaborative process. Developers need to work together as a team, coordinate with other teams (like design or marketing), and often interact directly with customers or users.



The developer persona



Understanding of user needs

Developers need to understand the needs and behaviors of the end-users of the software they are building. This helps them build software that is user-friendly and meets the needs of the users.



Aversion to marketing

It's fair to say no one likes being marketed to, but developers appear to have an aversion to traditional marketing strategies that may work with other audiences, so it's critical to engage them from a different perspective.

They need to be shown evidence, and every claim or point you make must be proven.

These characteristics can vary greatly between developers, and not all of them will have these traits to the same degree. However, these are a few broad tendencies that can help you distinguish developers from other roles.



"Developers make/build products and face countless challenges, and this process is time and energy consuming. To do their job well, they need the right tools. That's why they like marketing content that is honest and straightforward, useful and helpful documentation, and the chance to try out a product before deciding to buy it."

Mustapha Mekhatria, Community Manager at Highsoft

The developer persona

Developer expectations and pain points

When building your developer personas or thinking about who your audience is, you'll want to consider their expectations and pain points as well.

Here are a few of them:

Expectations

- Developers expect that the product being marketed to them is described with **technical accuracy**, so they understand how it works and what it can do (for instance, how to interact with its programmatic interface, such as an API).
- They want to know how the product solves their problems, so they expect you to provide relevant **use cases** that show the product's effectiveness in similar scenarios to their own.
- Developers also expect clear, transparent **pricing**, as they want to know what they're getting for their money, from maintenance to additional features.
- They want robust support and detailed **documentation**, which means having access to resources that can help them understand, implement, and troubleshoot the product.
- Developers expect their products to have a high level of **security**.
- They want it to be **scalable** as well, so that, as their projects grow and evolve, the product can keep up and meet their needs over time.
- Developers work with many different tools and tech, so they want products that are **compatible** with their tech stack.
- Many developers expect to be able to try the before before they buy it, which can be as a **free trial** or **demo**.

The developer persona

Pain points

- The field of software development is always evolving, with new languages, frameworks, and tools being introduced regularly. **Keeping up with tech changes** can be challenging.
- When short-term solutions are implemented instead of the best long-term solution, it can lead to a buildup of **technical debt** that can hinder future development.
- **Making code run efficiently** can be a complex task that requires a deep understanding of the language and system.
- Developers may find it challenging to **integrate products and systems**, especially when they're built by different teams or use different technologies.
- Being asked to complete work in an unrealistic time frame can be stressful and can lead to burnout, so **unrealistic deadlines** are a common pain point.
- Whether it's code that's not properly commented, or external documentation that's out of date or incomplete, developers often cite **poor documentation** as a major pain point.
- Developers often need to collaborate with other developers (many development teams are global), as well as with non-technical stakeholders. **Communication issues** can lead to misunderstandings.
- The **absence of appropriate** tools or having to work with outdated or inefficient tools can impede developers' productivity.

The developer persona

Action points when creating content for developers

Considering developer expectations and pain points, there are a few things to take into account when creating content that engages developers:

- Your content should be clear, accurate, detailed, and delivered in a variety of formats. It should also be distributed across multiple channels, since developers gather in communities across several social platforms, such as Reddit, Discord, Hacker News, etc.
- Maintain a factual and conversational tone that is both engaging and informative.
- Keep relevance, timeliness, credibility, consistency, and community in mind when planning and delivering your marketing campaigns.

But what types of content resonate with your developer audience? Here are a few popular ones:

- Tutorials and hands-on guides
- Documentation and API references
- Code samples and examples
- Blog posts and articles
- Video tutorials and webinars
- Community forums and Q&A platforms

The developer persona

Key developer attitudes

While developers aren't a hive mind, they do have a lot of things in common, including the fact that they:

- Trust hands-on experience over marketing claims
- Constantly upskill to stay up-to-date with modern technologies
- Are problem and use case driven
- Wish to stay current with latest technologies
- Are heavily influenced by peers



In regards to how developers think, **Deepak Kumar**, CEO at Doc-E-ai, also adds:

"Developers suffer from FOMO on emerging tech. Developers are mostly running late on their delivery. Developers always work in a flow. Hence, hate distractions. Developers hate fluffy language, as they are tasked to deliver a real product.

"I have never heard of a software project that was delivered on time with high quality. R&D folks mostly appear pensive, as they are always reconsidering about all the internal engineering trade-offs they did to deliver the software."

The developer persona



Liza Cichowski, Product Marketing Consultant at Last Word MKTG, has this to say about the developer audience – and how you can engage it:

“It can be hard for marketers to break through to an audience that doesn’t have much of an appetite for marketing. But software companies that don’t market to developers could be missing out. The U.S. Bureau of Labor Statistics projects that employment of software developers and other technologists will grow 22% by 2030. That’s a huge opportunity!

“To reach developers, the most important thing to keep in mind is that they are researchers. They like to seek out the information they need, analyze it and make a decision. As marketers, it’s our job to make this process easy for them. Here are a few suggestions:

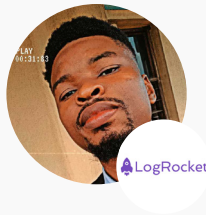
- Developers love to try before they buy. Give them the chance to test out your product for free and enable them to provide you with valuable insight.
- Developers know the difference between a must-have and nice-to-have product. Don’t use jargon or fluffy marketing language to communicate to them. Be very clear about the product’s value and how it will benefit them.
- Developers are uniquely positioned to understand both the industry landscape and their company’s needs. They are cross-functionally integrated throughout the organization so they know what’s going to be the best-fit solution for their colleagues. That’s why speaking to developers and creating personas are essential to any marketing strategy to reach them.”

Chapter 4

Benefits of developer personas

Benefits of developer personas

We've already brushed up on this, but let's look closely at the many benefits of creating developer personas – and why they're so important to guide your marketing messages, your content, and your overall campaigns.



"Creating developer personas is crucial to the success of your product or content. If you understand the developer mindset, you can efficiently position your product and skip past most marketing problems.

"The developer persona also helps you develop your product better. Understanding that the developer is focused more on solutions than aesthetics and fluff will help you properly align your goals with the requirements of your audience."

Fimber Elemuwa, Technical Author at Logrocket

Benefits of developer personas

Why should you build developer personas?

Understand your technical audience better

If you have developer personas, you can more easily keep your ideal customer in everyone's minds. You also get deep knowledge of developer pain points, as well as their goals, motivations, and behaviors.

Stand out from the crowd

Because personas allow you to create content that resonates with your audience, you can better reach out to devs, solve their problems, and stand out from your competitors.

Better use of resources

Creating developer personas helps you make the most of your resources. For example, your documentation is more focused and detailed, and you can better identify the right users for targeted testing too.

On top of this, you can better tailor your marketing strategies to your audience in a way that resonates with them as well.

Improved product development

Building developer personas helps you understand the needs of your audience – so, your product development is much more efficient, saving you time and money.

Benefits of developer personas

Improved user experience

You can also design a more intuitive and user-friendly experience that caters specifically to your developer audience.

Better communication

Personas help in crafting effective communication strategies that work well with your target market, which leads to improved marketing, sales, and support efforts. After all, with developer personas, you can better create products and messaging that are relevant to your target audience.

Informed decision making

Developer personas can guide product development decisions by highlighting the features that are most valuable to your users.

Better prioritization

When you have a clear understanding of who your audience is, their needs and pain points, their behaviors, traits, motivations, etc., you can prioritize features and bug fixes that are most important to them.

More empathy

With personas, you can foster empathy among your team, helping everyone from designers and developers to marketing and management understand who they're creating a product for.

Improved onboarding

By understanding the different types of developers who might be using your tools or platforms, you can create more effective onboarding experiences tailored to each persona. This can reduce the learning curve and encourage developers to start using your product.

Benefits of developer personas

Personas and developer communities

Understanding developers and getting into their mindset is crucial when you're interacting with communities, so you can make the most of them.

After all, there are many advantages of belonging to a developer community, both to its members and to the larger tech ecosystem. Developers form communities and engage with their peers in order to:

- Share knowledge
- Solve problems
- Network
- Contribute to open-source projects
- Collaborate with other developers
- Provide feedback to others
- Stay-up-to-date with the latest tech, trends, and best practices
- Influence product development
- Access a pool of resources

Benefits of developer personas

Forums, blogs, and Q&A platforms play a vital role in facilitating communication, collaboration, and learning within the developer community.

If you want to create effective marketing strategies, it's crucial you **identify and engage with relevant** developer communities and forums.

On top of this, high-quality **documentation** and tutorials are critical for attracting and retaining developers on your platform.

Also, leveraging **events and hackathons** can help drive community growth, increase brand awareness, and gather valuable feedback and insights.

Here are some recommendations for the **best tools you can use to engage** with developer communities, as well as **what they're best for**:

Tool	Best for
 Twitter	Updates, announcements, conversations, opinions
 StackOverflow	Technical discussion and thread, user forms
 YouTube	Video tutorials
 LinkedIn	Business-related posts, product announcements, major updates/launches
 Reddit	Opinionated, short conversations; Fun shares
 Discord	Tinkering; Group voice chats; Troubleshooting

Benefits of developer personas



"When creating developer personas, it is crucial to identify the appropriate communities, platforms and media to reach them. The developer media landscape is rich with platforms like Stack Overflow, Discord, GitHub, Slack, and Twitter.

"However, it is important to focus on the specific platforms that are most relevant to your goals. If your project is open-source, prioritizing the creation of a vibrant, active and inclusive community should be a key consideration.

"This involves targeting developers who actively participate in collaborative projects and enjoy contributing to open initiatives. On the other hand, if your tool is based on proprietary code, the emphasis should be on attracting targeted audiences to smaller, private communities.

Alice Malby, Lead Generation Manager at Front-Commerce

Who should be involved in creating your developer personas?

It'll probably vary, but try to think of everyone who'll have knowledge of your target audience.

Will your executive leadership understand your segment of the market? Your sales team?

What about marketing?

By compiling the knowledge of several individuals or teams, you can more easily build comprehensive developer personas that embody all the values, goals, motivations, behaviors, etc., of your product's users or buyers.

Chapter 5

The different types of developers

The different types of developers

As a developer marketer, your target market is not “all developers”, but a section of this audience. When you’re building your marketing plan, you need to keep your developer personas in mind!

Developer persona archetypes

There are several different types of developers – after all, they’re a varied bunch and no two developers are the same. However, they can share features and characteristics, so you can find patterns that help you understand them better.

The following aren’t hard-and-fast archetypes, but they’re useful to help you find the marketing strategies that work for your specific audience:

Hobbyist	Professional	Entrepreneur	Educator
			

The different types of developers



Hobbyist

People who engage in software, game, or web development, for example, as a hobby, instead of as a full-time job. These devs might be students, professionals in non-tech fields, retired individuals, or anyone else who has an interest in coding.

Hobbyist developers often:

- Learn and experiment
- Build personal projects
- Participate in communities
- Stay up-to-date with the latest tech

Many hobbyists are highly skilled and contribute significantly to various projects and communities.

The different types of developers



Professional

Individuals who engage in development (of games, products, apps, etc.) for their professional career and are, therefore, paid for the work they do. They're typically full-time, though part-time and freelance are also common.

You can find these developers in a range of industries, such as tech, healthcare, and finance.

Professional developers usually:

- Work on specific projects
- Have formal education or training (though many are also self-taught)
- Follow development processes
- Use a variety of tools and languages

When creating your personas, you may also want to consider the career stage your developer audience is at. For instance, are they new to the field, looking for a change, career-focused, or searching for work-life balance?

Keeping these in mind will help you understand your audience better.

The different types of developers



Entrepreneur

Entrepreneurs tend to combine their development skills with business activities. This means they use their technical knowledge to create software products, mobile apps, indie games, start-ups, and more.

They tend to:

- Build products from scratch with the intention of selling
- Start companies
- Wear multiple hats (including marketing)
- Build their own teams

Entrepreneur developers can make significant contributions to the tech industry and the broader economy, launching innovative products and businesses, creating jobs, and often pushing the boundaries of what's possible with technology.

The different types of developers



Educator

These developers teach and educate the next generation of devs, such as teaching them how to code. This can be anything from creating educational content to mentoring someone and promoting an inclusive learning environment.

On the other hand, you also have developer advocates, who serve as a bridge between a company (such as a tech company or an org that produces software or APIs) and its developer community.

They not only need a deep understanding of your company's products, but also strong communication and community-building skills.

They often:

- Promote a tech product or service
- Create educational content
- Participate in events, like meetups and hackathons
- Provide feedback to the company

The different types of developers



13 developer personas archetypes, by Zach Goldie

Zach Goldie is a positioning consultant helping B2B tech companies overhaul the underlying pitch for their campaigns.

He has a handy grid that showcases thirteen developer archetypes, **as well as the different levels of awareness customers can have about your product**, which further helps you understand where developers are in the user/buyer journey.

Happily Unaware Currently doesn't touch anything related to your product and hasn't yet contemplated doing so.	Dubious Newbie Is tempted by exploring new territory, but is wary about doing so.	Solution Unaware Wants the outcome you offer, but is unaware that those outcomes are achievable.	Enthusiastic Newbie Actively looking for a solution for the first time, often a free/OS starter option.
Happy Freebie user Doesn't know/care about the shortcomings with their OS or DIY option	Jaded Freebie user Frustrated by the cheap option, but doesn't think it's worth upgrading.	Upgrade unaware Wants a better tool but is unaware that options exist.	Upgrade seeker Actively looking for an upgrade, figuring out the best options.
Happy pro user Happily using a competing premium tool, unaware of any issues	Jaded pro user Frustrated by the pro option, but reluctant to migrate.	Migration unaware Moans about the tool but isn't aware that there are alternatives	Migration seeker Actively looking to switch providers, weighing up potential option.
	Jaded Veteran Tried similar tools in the past but was let down and gave up on the idea.		

The different types of developers

Developer roles

Developers can have a wide range of roles in the tech industry and beyond, depending on their skills, interests, and goals. Here's a list of some common roles devs can occupy:

- Front-end developer
- Back-end developer
- Full-stack developer
- Mobile developer
- Game developer
- DevOps developer
- Software developer
- Embedded developer
- CRM developer
- Graphics developer
- Security developer
- Machine learning engineer
- Blockchain developer
- System developer
- ML/Data Scientist

The different types of developers

Don Pinto, Sr. Principal Product Marketing Manager at Talend and Developer Marketing Alliance ambassador, further expands on some of the roles developers can choose:

“Front-End Web Developer: They focus on crafting captivating user interfaces and enhancing user experiences. They have a strong command of HTML, CSS, JavaScript, and various front-end frameworks. They prioritize experiences, interactivity, and seamless interactions.

“Back-End Web Developer: This persona specializes in server-side development, handling databases, APIs, and business logic. They are proficient in programming languages like Python, Java, Ruby, or PHP and are adept at architecting scalable and secure systems.

“System Developer: This persona focuses on system-level development, focusing on building robust and efficient software systems that interact directly with hardware, such as device drivers, embedded systems, or real-time applications. They have a deep understanding of operating systems, networks, and low-level programming languages like C or C++. They excel in optimizing performance, ensuring scalability, and handling complex infrastructure challenges.

“Full-Stack Developer: This persona possesses a well-rounded skill set, capable of working on both front-end and back-end development. They can handle end-to-end development, from creating intuitive interfaces to building robust server infrastructure.

“ML/Data Scientist: This persona thrives on data analysis, machine learning, and statistical modeling. They possess strong programming skills, along with expertise in tools like Python, R, or SQL. They use data-driven insights to make informed decisions and build intelligent systems.

“Mobile App Developer: This persona specializes in mobile application development for iOS or Android platforms. They are skilled in languages like Swift or Kotlin and have a deep understanding of mobile UI/UX design principles and app performance optimization.”

Chapter 6

Getting into the developer mindset

Getting into the developer mindset

Thinking like a developer is important for everyone working with developers, as well as building and selling products for this technical audience.

Your developer community is a great place to pay attention to what developers are doing and to act accordingly (i.e., to create a marketing strategy that resonates with your audience and leads to product adoption and sales).

Importance of thinking like a developer

Getting into the developer mindset helps you to:

- Understand your target market (their needs, pain points, behaviors, motivations, etc.)
- Speak their language
- Build credibility
- Create better marketing messages
- Inform your product development
- Communicate better with developers
- Build long-term relationships
- Understand a product's technical requirements

Getting into the developer mindset

How to get into the developer mindset

“Don’t make assumptions. Even as somebody who was a full-time software engineer and a full-time developer, I still find myself making assumptions about the type of documentation developers want to see or a particular programming language they probably want to hear about.

“I’m in dangerous territory, if that starts to happen. We should always be not only listening to the community, but also looking at the data that’s coming in, and use that to decide how we want to craft a message and things developers can use.”

[Liz Moy](#), Developer Advocate Lead at Evervault

Thinking like a developer is key to effectively selling products to developers. Here are a few things to consider as a starting point, so that you can better get into their mindset:

Understand developers’ challenges

Developers face a variety of challenges, from staying updated with the latest technologies to debugging complex code. Understanding these challenges can help you position your product as a solution.

Value quality and efficiency

Developers appreciate products that boost their efficiency and productivity. They value quality code, streamlined processes, and effective tools that help them do their jobs better.

Getting into the developer mindset

Speak developers' language

Familiarize yourself with technical terms and concepts that are relevant to the product you're selling. Developers will respect you more (and are more willing to listen to what you have to say) if they see that you understand their world.

Appreciate open-source and community

Many developers contribute to open-source projects and are part of various developer communities. They value transparency, collaboration, and collective problem-solving.

Think logically and structurally

Developers have to think logically and structurally to write code and solve problems. When selling to them, it helps to present your product in a clear, logical manner, outlining its features, benefits, and how it can solve their problems.

Focus on continual learning

Developers are lifelong learners who are always updating their skills to keep up with rapidly evolving technology. Highlight how your product can facilitate this continual learning.

Consider hands-on experience

Developers prefer to try out tools and technologies before committing to them. Offering a free trial or a demo can be a good way to attract them to your product.

Remember to be data-driven

Developers are often data-driven, so backing up your claims with data and real-world examples can be very persuasive.

Getting into the developer mindset

The 5 'Ws'

Liz Moy, Developer Advocate Lead at Evervault, talks about the **5 'W' questions you can ask to get you into the developer mindset:**

5 'W' questions		Tactics
Who?	Your developer or non-developer audience who can code or use a low-code tool. What are the things they like or the messages they respond well to?	• Surveys • NPS (net promoter score) • Developer community CRM (customer relationship management) • Developer feedback • Partnerships
What?	What do developers need to be successful and how can you help them solve their problems and meet their needs?	• Documentation • Swag • Hackathons • Frameworks • Integrations

Getting into the developer mindset

5 'W' questions		Tactics
When?	When, in the life cycle, does the developer use your product? Maybe they get told to use it or they want to experiment with it for fun. Ensure you've got content that targets your audience at different stages of the journey.	• 'When' might be tough to determine • Use developer evangelists and advocates for this
Where?	This can refer to geographical terms and also to where the developer is in their development or programming journey.	• Analytics • CRM • Surveys
Why?	The reasons why developers use your product. Maybe it saves them time, has features no other product has, or keeps their costs down.	• Meetups • Forums • Advocacy group

This should help you to put yourself in your audience's shoes.

Getting into the developer mindset



"When creating personas, people often think about the end result that devs are after. But, a lot of the time prospects already have a way to achieve the end result, whether it's tracking bugs, deploying code, managing APIs etc.

"Maybe they're doing things manually, fudging it with an OS tool or using a competing product. In those cases, the problem/solution you're solving isn't about the end result, but with overcoming the specific headaches in their current way of achieving that outcome.

"For example, an autoscaler isn't just solving app slowdowns since prospects are already fixing them manually. The pain point is the slowdown alerts in the middle of dinner.

"That's why I always start with defining the customer scenario, of how they're currently approaching the task and the awareness of any shortcomings."

Zach Goldie, Positioning Consultant

Chapter 7

The non-technical persona

The non-technical persona

Your target audience may not always be developers. Sometimes, it's a project manager or an engineer who hasn't coded in years, for example. So, understanding this non-technical persona can help you communicate about your product in a way that appeals to them.

What is a non-technical persona?

A non-technical persona is someone who doesn't have a deep understanding of, or even familiarity with, the technical aspects of a product. They may not have a background in technology or related fields, and may not be comfortable with technical jargon or complex explanations.

Examples of non-technical personas

- **Product manager:** responsible for guiding the success of a product and leading cross-functional teams.
- **UX/UI designer:** responsible for the user experience and interface design of software products (the role mainly requires skills in design and user psychology).
- **Technical writer:** this role involves creating documentation for technical products and, while the writer needs to understand the technology well enough to explain it, they don't necessarily need to know how to code or design systems.
- **Sales representative:** they need to understand the technology they're selling, but don't need to be able to create it.
- **Marketing specialist:** marketing professionals in tech companies need to understand the technology they're marketing, but their main skills are in areas like market research, copywriting, and strategic planning.
- **Project manager:** responsible for planning, executing, and overseeing projects to ensure they are completed in a timely fashion and within budget.

The non-technical persona

- **Business analyst:** responsible for bridging the gap between IT and the business using data analytics to assess processes, determine requirements, and deliver data-driven recommendations and reports to executives and stakeholders.
- **Data analyst:** primarily uses software to interpret data and identify trends, not to build or maintain technical systems.
- **Customer support:** these professionals handle customer inquiries, complaints, and technical issues, and don't need a deep understanding of the tech behind those products.
- **Community manager:** responsible for building, nurturing, and managing the community around a brand or product.
- **Operations manager:** oversees the production of goods and/or provision of services. They need to have a broad knowledge of many functions, including supply chain, quality control, and company infrastructure.

When do you need to market to a non-technical audience?

If you're marketing a new software product, you might have to target a non-technical persona – for example, a small business owner who needs the product for their business operations. They're primarily interested in how the software helps their business, instead of the intricacies of how the product is built or the specific technical features it may have.

By understanding this persona better, you can shape your marketing strategies in a way that's accessible to them. **The focus should be on benefits and solutions rather than technical details.**

The non-technical persona

How to market a technical product to a non-technical persona

As mentioned, this involves a strategic approach focused on the benefits and results of the product, rather than the technical details. The goal is to communicate the product's value in a way that is easily understandable, relatable, and relevant to the non-technical audience.

Here are some strategies:

- **Understand your audience** through market research.
- Highlight **benefits, not features**, as non-technical audiences are more interested in the benefits gained from using your product rather than its technical features.
- Use **simple, clear, and straightforward language** to make the product as easy to understand as possible.
- Use **visuals and stories** to simplify complex concepts, as and stories can make your product more relatable.
- Offer excellent **customer support**, since non-technical customers might need more support in understanding and using your product.
- Leverage **testimonials and case studies** to use as compelling proof of your product's value.
- Choose the **marketing channels** that your non-technical audience uses.

The non-technical persona

Further information: you can learn more about this topic in our podcast episode [“When is a developer not a developer?”](#), where we spoke to William Chia, Product Marketing Leader at Osano, about marketing to non-technical people.



Chapter 8

Creating developer personas

Creating developer personas

Creating effective developer personas requires a systematic approach and a good understanding of your target audience. The goal of building these personas is to help you and your team better understand your users and make decisions that meet their needs.

The more accurate and detailed your personas are, the more useful they'll be in achieving this goal.

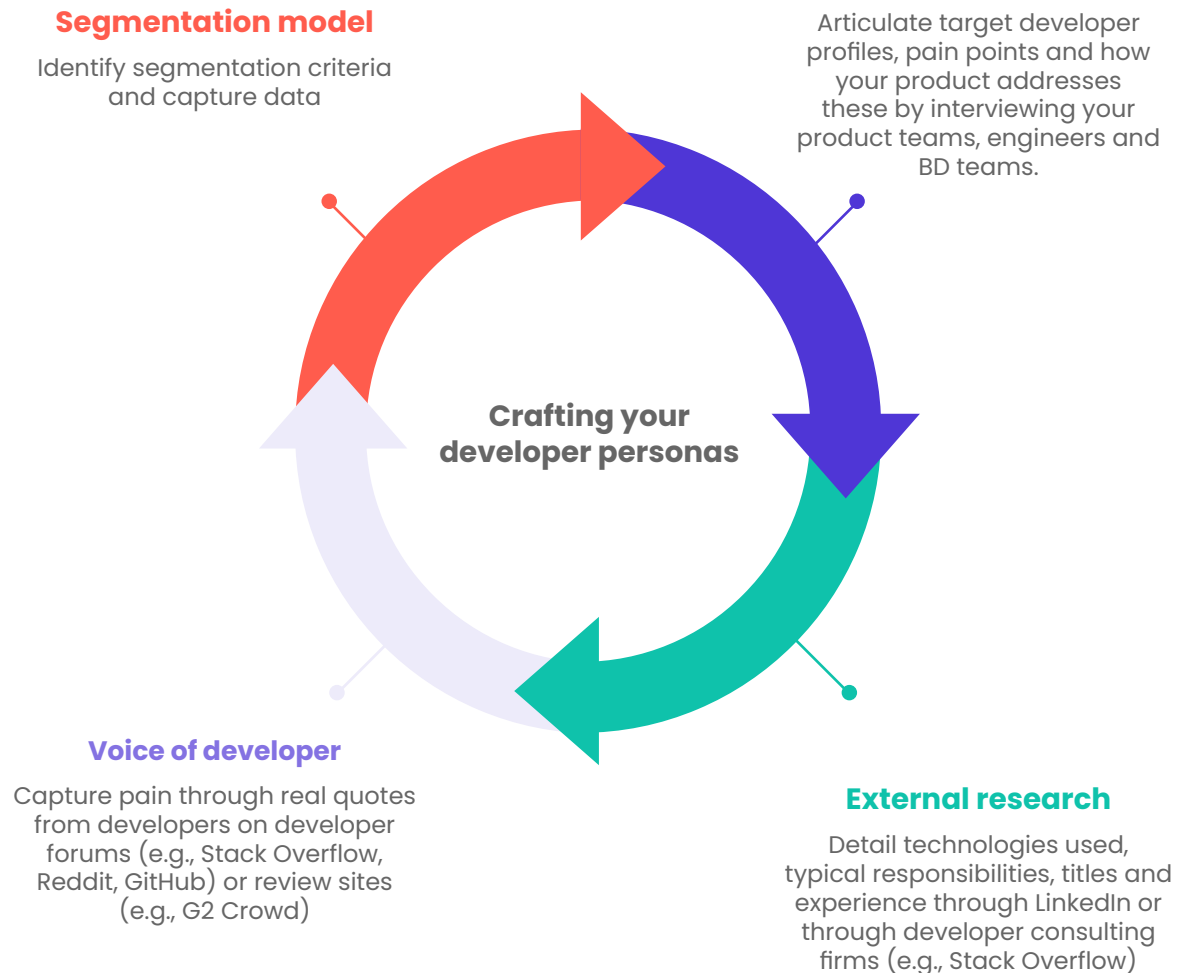
Crafting your developer personas

Creating developer personas can be a game-changer. They can guide your product development, shape your marketing strategies, and ultimately, help you build a product that truly resonates with your users.

But, where do you even start? Let's look at the different steps to craft your personas and what each of them entails.

technology they're marketing, but their main skills are in areas like market research, copywriting, and strategic planning.

Creating developer personas



Creating developer personas



1. Internal audit

Check your internal resources before searching externally. Your teams are a treasure trove of information, so make sure you talk to sales and product, for example.

First, define what you want to achieve from the audit. This could be identifying pain points or understanding the demographics of your user base.

You'll want to gather data through your **CRM**, your **analytics**, and **interviews** with your team members, in order to identify patterns, trends, and key takeaways.

Creating developer personas



2. External research

This step plays an important role in creating developer personas, as it allows you to gain a broader understanding of the industry, trends, and the overall landscape in which your customers operate.

This type of research also supplements the internal data you gathered. You can perform **industry** research, **competitor analysis**, and **market trends** research to stay on top of the latest tech and to anticipate future needs and changes.

On top of this, you can scour **social media** platforms, since they can offer you valuable insights into the challenges, needs, and preferences of developers.

What's more, **conferences**, **meetups**, and **hackathons** are also great opportunities to gather information about your audience, especially as you can more easily conduct interviews and surveys.

Other types of external research you can consider are:

- Industry publications
- Hiring trends
- Customer reviews and feedback

Creating developer personas



3. Voice of the developer

In this step, you want to capture the developers' perspective, which includes experiences, needs, and motivations. Don't skip this, as it can help you accurately represent the developers you're looking to depict.

You can capture the voice of the developer through different means, such as **direct feedback**, **user research**, and **observation** of behaviors and interactions. You can also **interview** developers, **survey** them, and participate in **forums/communities** like Reddit, GitHub, and Stack Overflow.

Make sure you understand the developer culture too, since developers have their own unique culture, full of humor, jargon, values, and norms. Understanding this culture helps you craft personas that feel authentic and relatable.

Creating developer personas



4. Segmentation model

Ensure you identify segmentation criteria as well. A segmentation model is used to categorize developers into different groups based on certain attributes (goals, needs, behavior patterns, etc.).

So, your model can look like:

- **Identify** the attributes you'll use to segment your audience based on data you've gathered.
- **Define** the segments, since each should represent a distinct group of developers with similar attributes. For example, all 'Senior Developers' may be part of the same segment.
- **Create** personas for each segment, which encapsulate developers' typical characteristics, behaviors, and needs.
- **Validate** the segments with further research, such as additional interviews or surveys.

You'll want to revisit and update your personas regularly to ensure they continue to accurately represent your audience.

Creating developer personas



"There are many ways to segment developers, such as expertise, motivation for developing, and relationship to your business (ex: customer vs partner). Such segments help with marketing strategy – how much nurturing is needed, what resources and programs are preferred, and how much to invest compared to value to business.

"Across developer segments, though, the mindset is consistent:

- Does this product or program solve my problems? Will it help me do my job better?
- Is this solution relatively easy to discover and get started with?
- Do resources exist to help me be successful?
- Can I trust that the company is invested in us being successful with this solution?
- Do I get value from this investment (recognition, credibility)?

"Even if segmented solutions exist, it's critical for the underlying values to be consistent. Developers seek both innovation and opportunity, which I place at the core of developer marketing."

Katie Miller, DMA Ambassador

Creating developer personas

“A persona should not just be a pure work of narrative fiction. When trying to understand an audience, it is helpful to have several representative examples in mind. It is easy to forget, but if you work in an organization that is building a technical product, you very likely have developers on your own teams who work in the industry you are focused on and are easy to reach out to.

“Make an effort to meet them and gather opinions on things you are working on. You may have a coworker from a previous role, a former classmate, or a colleague you connected with at a networking event. Reconnect with these folks as advisors who can fill in gaps and provide different points of view. There is no single developer mindset, but there are trends and your persona only has to go deep on the dimensions relevant for you making a decision. A front-end developer and a database administrator may have very different work outputs, but both may approach evaluating a technology to make a buy or build decision in the same exact way.

“These relationships are crucial to help you flush out a valuable persona that can help when making a decision. You’ve reached the goal when you can ask yourself ‘What would Brian do’ or ‘Would Katie read this?’ until you can intuitively answer those questions for yourself, literally reach out to your representative persona to get feedback. People are the critical ingredient to building a persona that helps segment the different types of developers you may encounter.

“As you build these relationships, you will gain a better understanding of how their specific role, their unique industry, their individual learning style, varying preferences for consuming information, and alternative views for looking at problems shape their work. As you get to know your persona surrogates better you can rely less on literally asking them questions and instead reference the empathy you’ve built in understanding how they think about things and captured in your persona definitions.”



Jason DeLancey, Sr. Manager of Developer Relations at Dolby

Creating developer personas

Here's an example of a developer persona:

Name: Sam the Senior Developer

Role: Senior Software Developer

Typical day: Sam spends most of his day coding, reviewing code, and collaborating with his team. He regularly participates in team meetings and contributes to decision-making processes about technical issues.

Technologies used: Sam primarily uses Java and works in an Agile environment. He's proficient in several IDEs but prefers IntelliJ IDEA. He uses Git for version control.

Challenges: Sam often struggles with integrating new technologies into existing systems. He finds it challenging to keep up with the fast-paced evolution of software development trends and technologies.

Goals: Sam wants to produce high-quality code and improve his technical skills. He also wants to help his team effectively implement new technologies.

Creating developer personas

Persona best practices

There are many ways to go about this process, but here are a few best practices to help you every step of the way:

- Don't skip the research. After all, no data = no persona.
- Don't get lost in useless details; only use information that's relevant to your goals and audience.
- Focus on just one or two persona goals, instead of filling out a template with several problems to solve. This can make it easier to stay on track and make decisions.
- Reconsider adding age. While developers come in all ages, this factor doesn't really matter when creating personas, since experience and motivations, for example, are more important.

Common mistakes to avoid

When crafting your developer personas, it's important to avoid:

- Making assumptions rather than using data and research.
- Overgeneralizing, since every developer is unique.
- Creating too many personas that can dilute your focus.
- Ignoring outliers that can provide valuable insights into opportunities.
- Not validating your personas.
- Not updating your developer personas.
- Bringing bias and assumptions to the process.



"Embracing and mastering the power of outside-in empathetic thinking is key to winning a market with a product. It begins by deeply understanding the mindset, needs and desires of buyers and power users. This profound connection with the audience lays the foundation for product success."

"When it comes to developer-facing products, this paradigm of thinking becomes even more crucial. Precise and comprehensive developer personas helps to understand, empathize and cater to the unique perspectives of the brilliant people behind the code, bridging the gap between a product innovation and market success"

Don Pinto, Lead Product Marketing Manager at Talend

Creating developer personas

Challenges to surpass

- You may also encounter a few challenges along the way, so here are a few common ones – and how to address them:

Limited access to developers

- Use online resources and participate in developer forums, Q&A sites like Stack Overflow, platforms like GitHub, and communities like the [Developer Marketing Alliance's Slack channel](#).
- Leverage internal resources by speaking to developers in your own org, and conducting interviews and surveys.

Overly broad or narrow segmentation

- If your segmentation is too narrow, you can try to identify what your segments have in common, allowing you to group them accordingly. If it's too broad, look for differences that could allow you to split them.
- Rather than basing your personas solely on job roles or experience levels, you may also want to consider factors like goals, motivations, preferred tools, and development methodologies.

Difficulty in validating personas

- Test and iterate by using your personas in a real-world setting (such as user testing or A/B testing) and gather feedback to refine them.
- Don't forget to also regularly request feedback. Make sure you're involving developers and other stakeholders in the validation process, as their input can help ensure that your personas accurately reflect the real world.

Creating developer personas

Lack of stakeholder buy-in

- Clearly articulate to your stakeholders how personas can help improve product development, marketing efforts, and overall business strategy.
- Include stakeholders in the process of creating and validating personas. Their involvement can help ensure buy-in.

Representing diversity

- Your research should include a diverse range of developers, whether from different industries, experience levels, or demographic backgrounds.
- If one persona can't adequately represent your audience's diversity, create multiple personas that reflect the range of your users.

Keeping personas up-to-date

- Set a schedule to regularly review and update your personas; this way, you can ensure they remain relevant as your product, market, and users evolve.
- Stay informed about changes and trends in the tech field that could affect your personas.

Ensuring usage of personas

- Make persona usage part of your standard processes for product development, marketing, and other areas.
- Provide training to educate your team about how to use personas and the benefits they provide.

Creating developer personas



"When creating developer personas, valuable insights can be gained from hours of in-depth research. However, the true worth lies in complementing research with direct conversations with developers. Engaging in meaningful dialogues unveils firsthand knowledge, experiences, and perspectives that go beyond secondary sources.

"The key to these conversations is preparing both specific and broad questions while giving the developers the space to freely express their wants and interests. Specific questions gather targeted information about technical preferences, pain points, and expectations, ensuring a comprehensive understanding.

"However, allowing developers to share their unique perspectives on developer relations and marketing is equally important. By listening attentively to their desires and interests in these areas, we can tailor our strategies and initiatives to align with their expectations, fostering authentic engagement and cultivating meaningful relationships."

Alice Malby, Lead Generation Manager at Front-Commerce

Creating developer personas

“Developers come in all shapes and sizes. The developer population is not a single homogeneous mass that can be expected to behave in a coherent way. For all its merits, diversity presents DevRel teams with a challenge. How do you reach out to a diverse set of people with a single message and expect it to appeal to most, if not all? Our answer is simple – you don’t.

“To maximize the return on your DevRel and developer marketing efforts, you need to segment the developer population into smaller groups, each as homogeneous as possible so that its members are highly likely to react in a similar – positive – way to a message especially tailored to them.

“Whilst this need is now broadly recognised amongst DevRel practitioners, many go about defining developer personas based on either arbitrary profile descriptions (e.g. by geography or student / professional status) or anecdotal evidence. At best, they may run a handful of interviews.

“We use hard data and solid methodology instead. Our survey data collected from more than 30,000 developers globally on an annual basis captures a representative view of the diversity of the developer population – whether you’re interested in the whole developer community or a subpopulation such as AI developers in N.America. We then let our unsupervised algorithms identify developer clusters that are indeed homogeneous in their needs and behaviors.

“Next, we ‘map’ the behavior and predict the most likely technology choices of each persona, i.e. we build models to predict persona technology choices. In this way, we obtain not an arbitrary set of profiles, but rather an actionable set of personas – built for the purpose of maximizing DevRel or even product launch ROI through targeted activities.

“In a nutshell: Anyone can come up with persona profile definitions. The trick is to come up with definitions that do the work you want them to do: To give you a clear picture of the distinct needs of the different developer clusters, and a set of recommended actions for each cluster that is designed to maximize the ROI of your DevRel investments.”



Viktorija Ignatavičiūtė, Sr. Community Lead at SlashData

Creating developer personas

Templates and frameworks

Lewko and Parton framework

The Developer Persona Canvas was developed by Caroline Lewko and James Parton; you can find it in their book, [Developer Relations: How to Build and Grow a Successful Developer Program](#).

This framework allows you to visualize developers' needs on a single piece of paper, which helps your DevRel (developer relations) efforts, your community management, and your developer marketing campaigns.

It also lets you compare several personas side by side, so you can better grasp the data.

Creating developer personas

Developer Persona Canvas

INDIVIDUAL Name Kyle Matthews  Location Atlanta, CA Company Software Agency Job Title Senior Backend Engineer SKILLS Languages Node JS, Python Platform AWS, Mongo DB Education Comp Sci Degree	AWARENESS 	PAIN POINTS - Need to stay at the cutting edge - Score creep - Time poor
	BUSINESS Scouting influence Lead Scout for new tech Purchasing Influence Key Influencer to CTO	USE CASES - eCommerce - Intergrations - Notifications - Delivery tracking
	PERSONAL DEVELOPMENT aws.training AWS-Reinvent university.mongodb.com MongoDB Live	
	COMMUNITY AWS User Group UK Serverless London Meetup London NodedJS User Group Londo NoSQL Meetup MongoDB.Local London	
		PRODUCTS Current OAuth API, OOBA API, Cloud Service Futre HDK

DISCOVER	EVALUATE	LEARN	BUILD	SCALE
Research Via - Google - StackOverflor - Peer Recommendations - Events	Reviews - Documentations - Demos - Reference Customers - Use Cases	- Code samples - Technical blogs - Tutorials - Forums - StackOverflow	- SDKs - GitHub - Forums - Support	- Use Cases - Account Management - Support

Creating developer personas

Developer persona template

You don't have to reinvent the wheel. There are many templates out there, such as this B2D (business-to-developer) buyer persona template you can use to create your personas, which is included in the Developer Marketing Alliance membership (you'll access many more templates by becoming a member).

B2D buyer persona example

Buyers could be anyone. In some businesses, it's the Director of Software Engineering; in others, the Chief Technology Officer; and, in others again, the VP of Engineering. It's on you to first find who that person is and then you can build out your buyer persona - [head here](#) for a blank template to start working on.

Remember: the characteristics are interchangeable. Depending on what's most relevant for your business, you may want to remove or add additional fields. We'd recommend a maximum of 10 fields per persona and this template is designed to provide inspiration for what those fields might be/look like.

JANE DOE

Bio	Jane's the decision-maker of technology for her company. She's all about balancing price and value. Her head's always in new technology and she has a lot of pressure from above to keep the business not only up to date but also ahead of the competition.
Role in the buying process	Jane has the final say on all large, recurring purchases. She oversees and develops technology for customers, vendors, and company staff. Jane also liaises with the CFO to ensure everything is within budget.
Background	Job: Chief Technical Officer (CTO) Reports to: CEO No. of employees: 6

Conclusion

In today's world of every-changing technology, understanding your developer audience is critical to building products that resonate with them.

Personas are a great tool in this process, since they offer a clear, detailed, and empathetic understanding of your target market – allowing you to tailor your products and messaging to developers.

Remember that developer personas are not static; they should evolve as your users and the industry change.

This playbook covered everything, from the developer mindset to the steps needed to craft powerful personas. Join the [Developer Marketing Alliance](#) to stay up-to-date with future playbook and report releases, and connect with your peers in our [Slack community](#).

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A special thank you

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Teresa is our Senior Copywriter here at the Developer Marketing Alliance. She has a background in digital marketing and has been creating content for companies across many different industries over the years. She's open to feedback and would be thrilled to hear from you!



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Graphic Designer

Meet Lucy, our talented Graphic Designer who specializes in fulfilling all our design needs. She's in charge of creating the layout and incorporating visual elements in this report and is always eager to hear your thoughts on her designs!

And a huge thank you to all our contributors and everyone who helped us create this playbook:



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